

Data Analysis Report

Cafe Sale Analysis Report

Showing how many missing(NaN) vaule are in each col/row

Cols	Missing Values
Transaction ID	0
Item	333
Quantity	138
Price Per Unit	179
Total Spent	173
Payment Method	2579
Location	3265
Transaction Date	159

Total data before cleaning is 10000 rows and 7 columns

I convert Total Spent,Quantity and price per unit to numeric data using to_numeric() method,which set all text to NaN values(including 'ERROR' and 'UNKNOWN').
Used to_datetime() convert Transaction date to dates,which remove all false value.
The dataset contains 3 numeric columns,4 categorical columns, and 1 date column.

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After Cleaning the Data

Each Item price in the cafe

Item	Price Per Unit
Coffee	2.0
Cake	3.0
Cookie	1.0
Salad	5.0
Tea	1.5
Smoothie	4.0
Sandwich	4.5
Juice	3.0

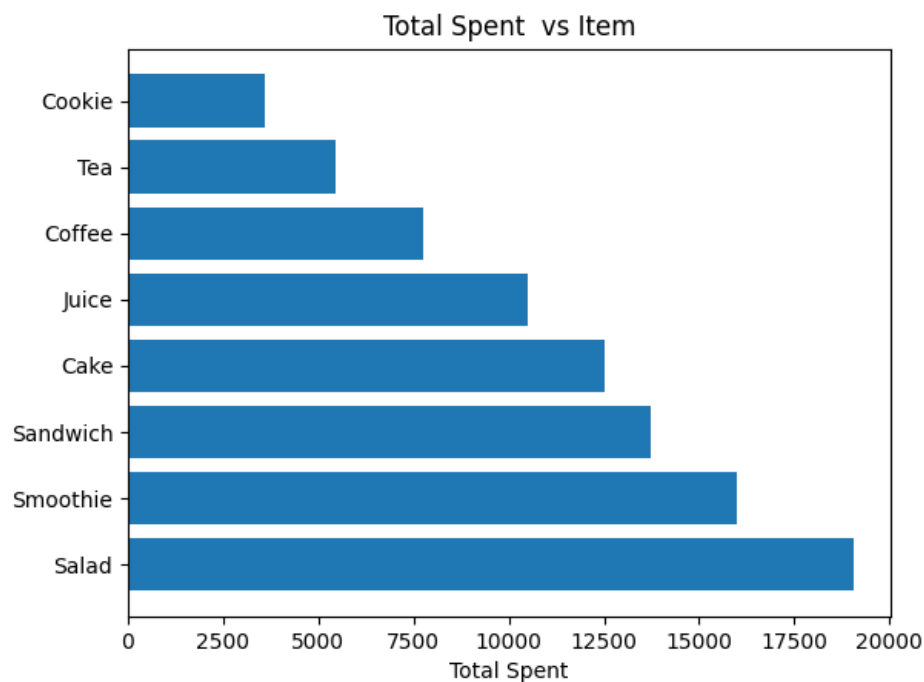
Total Clean Revenue: R88610.50

Recovered missing numeric values using mathematical relationships between Quantity, Price Per Unit, and Total Spent.
Used mapping dictionaries to restore missing Item names and prices.
For categorical or text data i used mode and 'other' to replace all the missing data.
Removed extreme outlier values using quantile(0.975).
All 7 columes are still in the data

Total data after cleaning 9926
About 74 rows have been removed

Data Analysis Report

Revenue by Item

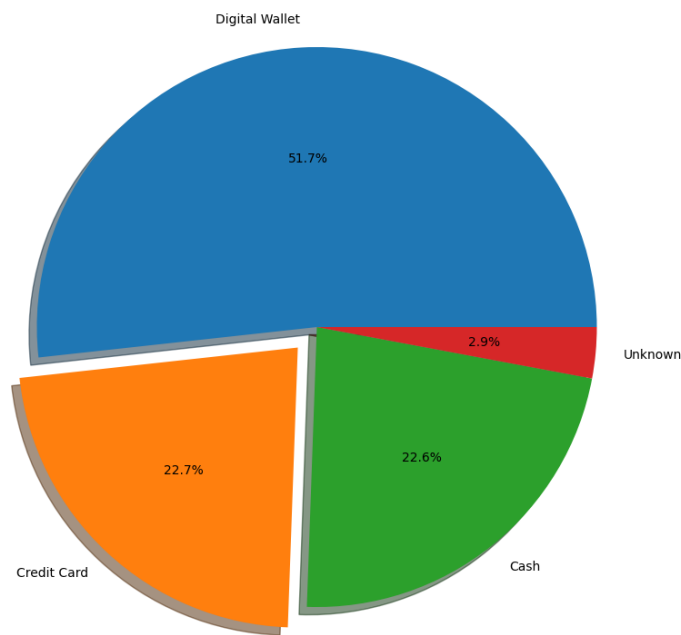


This chart shows the total revenue generated by each product sold in the cafe.

The analysis helps identify high-performing items that contribute most to total sales. Products with lower revenue may require marketing improvements or pricing adjustments.

The chart also provides insight into customer purchasing behavior and popular menu choices.

Payment Method Breakdown

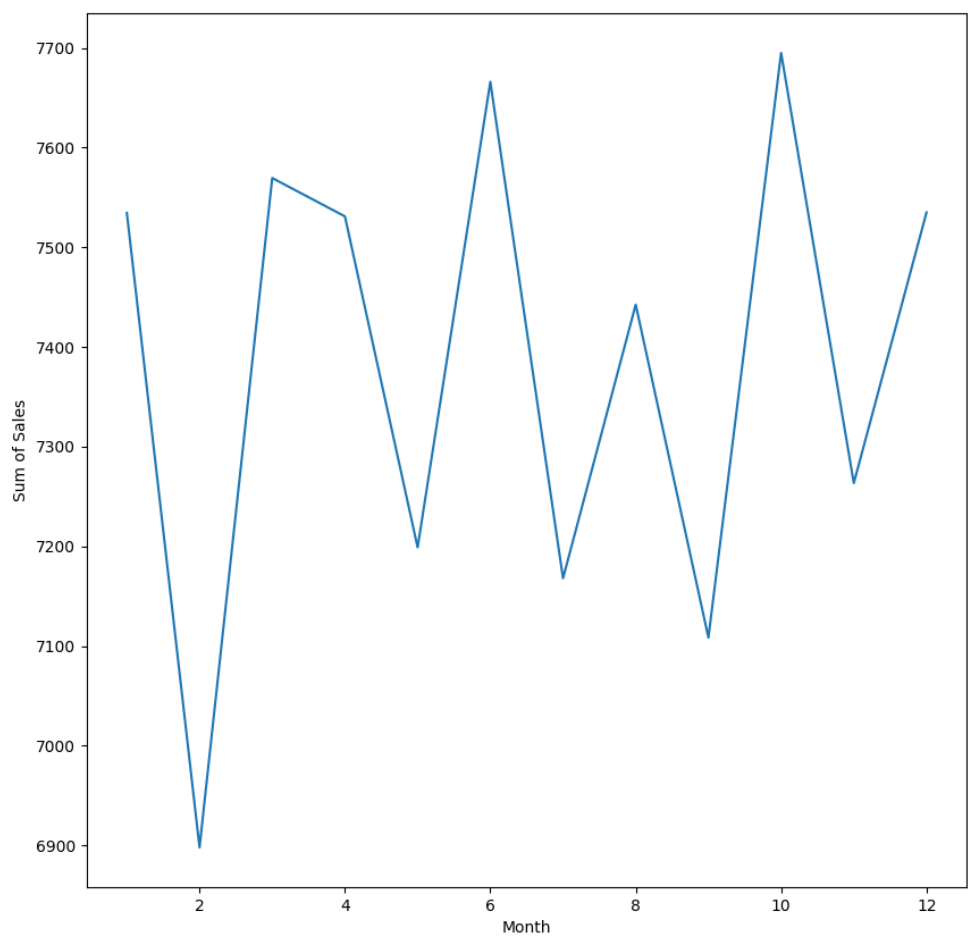


This pie chart illustrates the distribution of customer payment methods used during sales.

The results help understand customer payment preferences and transaction behavior.

Card and digital payments can indicate higher adoption of cashless transactions, while cash payments may reflect customer accessibility and convenience preferences.

Monthly Sales Trend



This line graph displays monthly sales trends based on total revenue over time.

The visualization helps identify periods of high and low sales activity within the cafe.

Sales trends are useful for forecasting, inventory planning, staffing decisions, and identifying seasonal customer behavior.